

LIMA, Peru

WMO: 846280

Lat: 12.022S

Lon: 77.114W

Elev: 113

StdP: 14.64

Time Zone: -5.00 (W05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	57.2	58.2	53.3	60.8	60.3	53.7	61.7	60.3	17.4	65.7	16.0	64.8	4.3	180	0.337

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	11.1	84.1	73.4	82.3	72.5	80.6	71.8	74.7	81.1	73.6	80.0	72.6	78.8	10.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.2	119.5	78.9	71.5	116.7	78.5	70.1	111.2	77.0	38.3	81.0	37.2	80.2	36.4	79.3	78.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.7	15.3	14.0	DB	56.6	87.0	2.5	2.4	54.8	88.7	53.4	90.1	52.0	91.4	50.2	93.1	(4)
(5)			WB	54.9	74.8	2.4	1.6	53.2	75.9	51.8	76.9	50.4	77.7	48.7	78.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	68.1	74.0	75.5	74.4	70.6	67.1	64.7	63.6	62.7	63.1	64.5	67.0	70.4	(6)
(7)		DBStd	5.24	2.65	2.55	2.83	2.84	3.15	3.43	3.07	2.49	2.46	2.23	2.28	2.47	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	318	0	0	0	0	14	46	65	82	68	36	7	0	(9)
(10)		CDD50	6606	744	714	756	619	532	441	422	393	393	449	510	633	(10)
(11)		CDD65	1448	279	294	291	169	80	37	22	10	11	20	67	168	(11)
(12)		CDH74	4215	978	1355	1173	328	76	29	17	6	6	3	23	221	(12)
(13)		CDH80	468	95	181	158	21	3	1	0	0	0	0	0	9	(13)
(14)	Wind	WSAvg	7.3	8.7	7.9	7.2	6.6	6.3	6.0	6.4	6.7	7.3	7.5	8.0	8.6	(14)
(15)	Precipitation	PrecAvg	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1	0.0	0.0	0.0	0.0	(15)
(16)		PrecMax	0.6	0.1	0.1	0.2	0.0	0.1	0.1	0.2	0.2	0.2	0.0	0.1	0.1	(16)
(17)		PrecMin	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	85.9	86.4	86.2	82.6	79.1	77.4	76.7	74.8	74.8	73.7	76.9	81.0	(19)
(20)			MCWB	74.8	74.8	74.3	71.6	69.2	68.3	67.4	66.9	66.8	65.4	67.6	71.5	(20)
(21)		2%	DB	82.4	84.2	83.9	79.2	75.6	73.6	71.8	70.1	70.2	71.4	73.7	78.2	(21)
(22)			MCWB	72.5	73.3	73.3	70.3	67.6	66.2	65.2	64.8	64.3	64.2	65.8	69.7	(22)
(23)		5%	DB	80.5	82.3	82.0	77.3	73.6	71.5	69.8	67.9	68.2	69.6	72.0	76.6	(23)
(24)			MCWB	71.7	72.7	72.4	69.4	66.7	65.8	63.9	62.3	62.7	63.2	65.0	68.7	(24)
(25)		10%	DB	78.8	80.5	79.7	75.5	71.8	69.5	67.7	65.9	66.4	68.1	71.2	75.0	(25)
(26)			MCWB	71.0	72.1	71.4	68.6	65.9	64.2	62.2	60.7	61.6	62.4	64.7	68.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.6	76.3	75.4	72.7	69.9	68.8	67.5	67.8	67.2	65.9	68.1	72.1	(27)
(28)			MCD B	82.6	83.7	83.2	80.4	77.8	75.3	75.0	72.5	73.3	72.3	74.8	78.6	(28)
(29)		2%	WB	74.1	74.9	74.2	71.2	68.3	67.3	66.0	65.4	65.4	64.5	66.8	70.7	(29)
(30)			MCD B	80.3	80.7	81.4	77.3	74.4	73.2	71.6	70.7	70.0	69.7	72.6	76.7	(30)
(31)		5%	WB	72.8	73.9	73.0	70.1	67.4	65.7	63.8	62.1	63.0	63.5	65.7	69.3	(31)
(32)			MCD B	78.7	79.9	79.5	75.6	72.7	70.8	68.9	66.5	67.6	68.4	71.0	75.2	(32)
(33)		10%	WB	71.6	72.8	72.0	69.2	66.4	64.4	62.4	60.8	61.4	62.8	64.9	68.2	(33)
(34)			MCD B	77.8	78.8	78.3	74.4	71.1	68.9	66.5	64.6	65.4	67.3	70.0	73.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.0	11.1	11.3	10.3	8.3	6.1	5.6	5.9	6.7	7.6	8.3	9.1	(35)
(36)			MCD BR	11.5	12.2	12.9	11.9	10.7	8.4	7.9	7.8	8.4	8.9	9.3	10.5	(36)
(37)			MCWBR	4.4	4.6	5.0	4.9	4.7	3.9	3.6	3.9	3.8	4.1	4.0	4.5	(37)
(38)		5% WB	MCD BR	10.8	11.2	12.1	11.0	10.1	8.0	7.1	6.7	7.9	8.5	8.9	9.9	(38)
(39)			MCWBR	4.1	4.3	4.9	4.7	4.6	3.8	3.5	3.6	3.9	4.1	4.1	4.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.499	0.490	0.510	0.502	0.448	0.448	0.426	0.449	0.535	0.486	0.479	0.488	(40)	
(41)		taud	2.186	2.228	2.168	2.162	2.282	2.266	2.344	2.259	1.999	2.170	2.201	2.199	(41)	
(42)		Ebn at Noon	271	272	261	252	256	251	260	263	250	269	274	273	(42)	
(43)		Edh at Noon	50	48	49	47	40	39	37	42	58	50	49	49	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1953	1901	1866	1802	1529	1228	1191	1337	1526	1745	1913	1880	(44)	
(45)		RadStd	145	85	71	88	128	93	72	71	106	95	159	161	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+220
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page